Team Contribution Journal: Preprocessing & Data Hygiene Pipeline

Project Title: Clinical Stroke Risk Classification Pipeline

Repository: Stroke-Exploratory-Analysis

Dataset Source: Stroke Prediction Dataset (Kaggle)

Arinze N Okafor (Team Lead)

Role: Project Lead, Architecture Design & Pipeline Engineering

- Repository Architecture & Version Control:

I initialized and structured our centralized GitHub project repository (Stroke-Exploratory-Analysis), set up our active development branch (Alpha), and defined clean commit and file-naming standards for the team. I also authored the primary dataset source documentation (dataset_reference) linking directly to the Kaggle raw data.

- Technical Proposal Deliverable:

I authored, formatted, and exported our formal One-Page Dataset Proposal (Dataset_Proposal.pdf), summarizing the clinical motivation, target feature taxonomy, row-count compliance, and the strict zero-leakage strategy we committed to follow.

- Pipeline Engineering & Leakage Isolation:

I built and executed the end-to-end Python preprocessing pipeline (stroke_preprocessing.ipynb) in Google Colab, ensuring the workflow runs cleanly from top to bottom via Restart & Run All. I

enforced an immediate stratified 70/30 train-test split (3,576 training rows, 1,533 test rows) prior to any calculations to ensure complete statistical isolation of our holdout set.

- Data Hygiene & Transformation Design:

I designed and executed all feature transformations strictly on the training partition: I dropped the non-predictive id key, pruned the single 'Other' entry in gender to avoid sparse dummy artifacts, and computed the training median BMI (27.90) to impute missing values across both splits. I applied One-Hot Encoding (drop_first=True) with schema alignment to guarantee matching dimensions, and standardized age, avg_glucose_level, and bmi using StandardScaler fitted solely on training observations.

- Class Balancing & Diagnostic Suites:

To address our ~19:1 clinical class disparity (3,402 non-stroke vs. 174 stroke cases in training), I implemented SMOTE exclusively on the training features, equalizing training classes to 3,402 each while leaving the test set authentic and untouched. I built the Matplotlib and Seaborn visual comparison suite displaying missingness profiles, continuous skewness, and class topologies before and after preprocessing.

- Deliverable Integration:

I exported our final processed CSV files (cleaned_train_stroke_data.csv, cleaned_test_stroke_data.csv) directly to our GitHub repository and consolidated all individual contributions into this final submission deliverable.

Hamza Ajmal

Role: Data Preprocessing, Distribution Normalization & Pipeline Verification

- Dataset Selection and Setup:

I assisted in evaluating the instructor-approved Stroke Prediction Dataset from Kaggle, verified that its 5,110 rows comfortably satisfied the 2,000-row minimum requirement, and worked directly with the raw CSV in Python rather than manually modifying the source file.

- Notebook Execution & Exploratory Analysis:

I prepared and ran the Jupyter Notebook workflow (Restart & Run All) to inspect raw feature distributions, verify our 70/30 split, perform training-only exploratory data analysis, generate diagnostic before/after distributions using Matplotlib, and log intermediate sample outputs during execution.

- Feature Normalization & Interaction Modeling:

I constructed an alternative scikit-learn preprocessing pipeline fitted strictly on training data: I evaluated Yeo-Johnson power transformations (reducing skewness in glucose from 1.58 to 0.08, and BMI from 1.04 to 0.00) followed by standard scaling. I implemented binary 0/1 encoding for ever_married and Residence_type, evaluated one-hot encoding across remaining categorical variables while retaining the 'Unknown' smoking category as an explicit category, and engineered an age–hypertension interaction feature.

- Outputs and Verification:

I audited pipeline outputs recording our original dataset shape ($5{,}110 \times 12$), training and test partition sizes ($3{,}576$ and $1{,}533$), training BMI missingness (148 missing values in train), and class distributions before and after training-only oversampling (3,402/174 before; 3,402/3,402

after). I contributed these technical metrics directly to our team reflection and documentation deliverables.

Jonathan Willturner

Role: Data Quality Assurance, Statistical Auditing & Evaluation Framework

- Data Integrity & Boundary Audits:

I conducted an independent quality assurance audit on our exported feature matrices (cleaned_train_stroke_data.csv and cleaned_test_stroke_data.csv). I verified that zero null values remained across all columns post-imputation and confirmed that our continuous features conformed to standard scaling properties ($\mu \approx 0, \sigma \approx 1$).

- Schema & Column Alignment Verification:

I audited the one-hot encoded dummy columns across both partitions, confirming that our categorical alignment routines successfully matched all 15 final feature columns and prevented zero-variance feature artifacts between splits.

- Evaluation Framework & Metric Specification:

I developed our downstream evaluation strategy for the upcoming modeling phase. I documented why raw classification accuracy must be discarded in this imbalanced medical context (~95.1% naive baseline) in favor of Recall (Sensitivity) to minimize life-threatening False Negatives, alongside Precision-Recall AUC (PR-AUC) over standard ROC-AUC.

- Documentation & Reproducibility:

I assisted in drafting our team reflection journal and reviewed our cross-validation plan to ensure future hyperparameter tuning embeds SMOTE inside an imblearn pipeline, preventing cross-fold validation leakage during model training.